

House Price Prediction Using Machine Learning

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on property characteristics such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and additional amenities.

Methodology

The Housing Prices Dataset from Kaggle was used for this analysis. The dataset contained 545 records and 13 features. Data exploration revealed that there were no missing values or duplicate records. Categorical variables were converted into numerical form using binary encoding and one-hot encoding techniques.

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio.

Results

Linear Regression:

- MAE: 970,043.40
- RMSE: 1,324,506.96
- R^2 Score: 0.6529

Random Forest Regressor:

- MAE: 1,014,947.32
- RMSE: 1,399,769.44
- R^2 Score: 0.6124

The Linear Regression model achieved the best performance with an R^2 score of approximately 65.3%.

Key Insights

The most influential features affecting house prices were bathrooms, air conditioning, hot water heating, preferred area, and furnishing status. Houses with better amenities and desirable locations generally had higher market values.

Conclusion

The project successfully demonstrated how machine learning can be used to estimate house prices. Linear Regression provided the most accurate predictions for this dataset and can serve as a useful tool for supporting real estate pricing decisions.

Additionally, [advanced_analysis.ipynb](#) was created to include extended exploratory analysis, feature correlation studies, model persistence using Pickle, and prediction demonstrations beyond the core project requirements.